

CCP-CNS2026 — Computer Networks

Problem 1: Multi-Site Enterprise Network

GROUP CONTRACT & TEAM CHARTER

Instructor	Sir Muhammad Fahad Irshad
Course	Computer Network Lab
Problem Assigned	Problem 1 — Multi-Site Enterprise
Team Size	4 members
Project Duration	5 Weeks
Date of Charter	May 2026
GitHub Repository	https://github.com/ZohaibAhmad260/CCP-CNS2026-Problem1
Section	V9

1. TEAM MEMBERS & ROLES

#	Name	Role	Primary Responsibilities
1	Zohaib Ahmad (F2024266251)	Project Manager & Network Architect	Team lead, topology design, IP addressing plan, VLAN & routing configuration, meeting scheduling, GitHub management, weekly status reports
2	M. Efan Amir(F20242661144)	Security Engineer	ACL configuration, DHCP snooping, attack simulation from Kali Linux, DoS execution, post-attack mitigation, incident response
3	Muhammad Abdullah (F2024266337)	Automation Lead & Testing/QA	GitHub repository setup & maintenance, Python scripts, Wireshark PCAP capture, traffic analysis, ping test matrix, performance testing with iPerf
4	Tayyab Hassan (F2024266511)	Documentation Lead	Weekly reports, final report compilation (15-20 pages), network diagrams in draw.io, video script, presentation slides, peer evaluation forms

2. PROJECT OBJECTIVES

This team is assigned Problem 1: Multi-Site Enterprise. The objective is to simulate a secure company network for Apex Manufacturing with the following goals:

- Design and implement a multi-VLAN network (Sales, Engineering, Admin) using GNS3
- Configure a Cisco router using router-on-a-stick for inter-VLAN routing
- Implement DHCP server on router for all three VLANs
- Deploy an Ubuntu web server in the Admin VLAN

- Implement DHCP snooping and ACLs to block Sales from accessing Admin
- Simulate a DoS attack from Kali Linux and capture traffic with Wireshark
- Mitigate the attack using ACL rules and document the full incident

3. COMMUNICATION & MEETING PLAN

Meeting Frequency	Minimum twice per week (flexible as needed)
Meeting Platform	WhatsApp group call / Discord / In-person
Primary Communication	WhatsApp group chat for all members
File Sharing	GitHub repository (all submissions pushed here)
Response Time	All messages to be acknowledged within 12 hours

4. FIVE-WEEK SCHEDULE OVERVIEW

Week	Theme	Key Deliverables
Week 1	Planning & Design	Team charter, topology diagram, IP plan, GitHub setup, risk matrix, resource acquisition
Week 2	GNS3 Implementation	GNS3 project file, all device configs, ping test matrix, VLAN screenshots, SSH proof
Week 3	Security & Attack	ACL configs, DHCP snooping, DoS attack script, PCAP files, mitigation proof, attack video
Week 4	Integration & Testing	Final GNS3 project, NAT config, performance baseline, full test matrix with pass/fail
Week 5	Final Submission	Final report (15-20 pages), demo video (8-10 min), presentation slides, peer evaluations

5. TEAM AGREEMENTS & CODE OF CONDUCT

5.1 Work Ethics

- Every member will complete their assigned role tasks on time
- No member will submit work they did not personally contribute to
- All code, configs, and documents must be pushed to GitHub — no personal storage

5.2 Late Work Policy

- If a member cannot complete their task, they must inform the at least 24 hours before the deadline
- The instructor's policy applies: 10% deduction per day, maximum 50%
- Internally, repeated lateness (2+ times) will be reported to the instructor

5.3 GitHub Usage Rules

- All deliverables must be pushed to the team GitHub repository before submission
- Commit messages must be descriptive (e.g., 'Added Week 1 IP addressing plan')
- Muhammad Abdullah (Automation Lead) is the repository admin

5.4 Conflict Resolution

- Disagreements are first discussed in the group chat
- Personal conflicts are kept out of project work

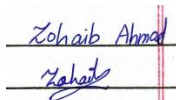
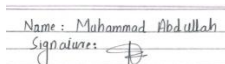
6. GRADING AWARENESS

All team members have read and understood the grading structure:

Component	Team Grade Weight	Points
Week 1 Deliverables	—	15 pts
Week 2 Deliverables	—	20 pts
Week 3 Deliverables	—	25 pts
Week 4 Deliverables	—	20 pts
Week 5 Deliverables	—	20 pts
Final Grade Formula	70% Team + 30% Individual	100 pts

7. SIGNATURES & AGREEMENT

By signing below, each team member confirms that they have read, understood, and agree to all terms in this Group Contract & Team Charter.

Full Name	Role	Signature
Zohaib Ahmad	PM & Network Architect	
M.Efan Amir	Security Engineer	
Muhammad Abdullah	Automation Lead & QA	
Tayyab Hassan	Documentation Lead	